

Homework 1, Convex Optimization

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1 Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ such that $|f(x)| \leq \|x\|^2$ for all x . Show that f is Fréchet differentiable at 0.

Proof: If f is Fréchet differentiable at $x = 0$, then there is a linear map l such that

$$\lim_{\|h\| \rightarrow 0} \frac{|f(0+h) - f(0) - l(h)|}{\|h\|} = 0$$

Taking $l := 0$ and since $-\|h\|^2 \leq f(h) \leq \|h\|^2$, we have that $f(0) = 0$ and

$$\frac{|f(0+h) - f(0) - l(h)|}{\|h\|} = \frac{|f(h) - 0 - 0|}{\|h\|} = \frac{|f(h)|}{\|h\|}$$

Since $|f(h)| \leq \|h\|^2$, we have that

$$\frac{|f(h)|}{\|h\|} \leq \frac{\|h\|^2}{\|h\|} = \|h\| \rightarrow 0 \text{ as } \|h\| \rightarrow 0$$

Therefore, f is Fréchet differentiable at 0. $\square$

2 Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined as

$$f(x) = \frac{xy^2}{x^2 + y^4}$$

if $(x, y) \neq (0, 0)$ and $f(0, 0) = 0$.

(a) Show that f has directional derivatives at $(0, 0)$ in all direction d .

Proof: Let $d = (d_1, d_2)$, then

$$\begin{aligned}
\frac{f(0+td) - f(0)}{t} &= \frac{((td_1)(td_2)^2)/((td_1)^2 + (td_2)^4)}{t} \\
&= \frac{td_1 t^2 d_2^2}{t(t^2 d_1^2 + t^4 d_2^4)} \\
&= \frac{t^3 d_1 d_2^2}{t^3(d_1^2 + t^2 d_2^4)} \\
&= \frac{d_1 d_2^2}{d_1^2 + t^2 d_2^4} \\
&\rightarrow \frac{d_1 d_2^2}{d_1^2} = \frac{d_2^2}{d_1} \text{ as } t \rightarrow 0 \text{ when } d_1 \neq 0
\end{aligned}$$

When $d_1 = 0$, we have that

$$\begin{aligned}
\frac{d_1 d_2}{d_1^2 + t^2 d_2^4} &= \frac{0}{0 + t^2 d_2^4} \\
&\rightarrow 0 \text{ as } t \rightarrow 0
\end{aligned}$$

Therefore, f has directional derivatives at $(0,0)$ in all directions d . $\square$

(b) Show that f is not Gâteaux differentiable in the origin.

Proof: In the previous point we showed that f has directional derivatives at $(0,0)$ in all directions d and the directional derivative is given by

$$f'(0, d) = \begin{cases} \frac{d_2}{d_1} & d_1 \neq 0 \\ 0 & d_1 = 0 \end{cases}$$

But this function is not linear in d because $1 = f'(0, (1, 1)) \neq f'(0, (1, 0)) + f'(0, (0, 1)) = 0 + 0 = 0$. Therefore, f is not Gâteaux differentiable at $(0,0)$. $\square$

(c) Show that f is not continuous at $(0,0)$.

Proof: Take the limit as $(x, y) \rightarrow (0,0)$ along the path $y^2 = x$, then

$$\lim_{y \rightarrow 0} f(y^2, y) = \lim_{y \rightarrow 0} \frac{y^2 y^2}{y^4 + y^4} = \lim_{y \rightarrow 0} \frac{y^4}{2y^4} = \frac{1}{2} \neq 0 = f(0,0)$$

Therefore, f is not continuous at $(0,0)$. $\square$

3 Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined as

$$f(x) = \frac{2y \exp(-x^{-2})}{y^2 + \exp(-2x^{-2})}$$

if $x \neq 0$ and $f(0,0) = 0$. Show that f is Gâteaux differentiable at the origin but not continuous.

Proof: Let $d = (d_1, d_2)$, then

$$\begin{aligned}
\frac{f(0+td) - f(0)}{t} &= \frac{2(td_2) \exp(-(td_1)^{-2})}{t((td_2)^2 + \exp(-2(td_1)^{-2}))} \\
&= \frac{2d_2 \exp(-(td_1)^{-2})}{t^2 d_2^2 + \exp(-2(td_1)^{-2})}
\end{aligned}$$

Since $\exp(-(td_1)^{-2})$ is $o(1)$ as well as $\exp(-2(td_1)^{-2})$, we have that

$$\begin{aligned} &= \frac{2d_2 \exp(-(td_1)^{-2})}{t^2 d_2^2 + \exp(-2(td_1)^{-2})} \\ &= \frac{o(1)}{t^2 d_2^2 + o(1)} \\ &\rightarrow 0 \text{ as } t \rightarrow 0 \end{aligned}$$

Therefore, the directional derivative exists in all directions d and is linear in d at $(0,0)$, so f is Gâteaux differentiable at $(0,0)$. To see that f is not continuous at $(0,0)$, take the limit as $(x,y) \rightarrow (0,0)$ along the path $x^2 = -1/\ln(y)$, then

$$\begin{aligned} \lim_{y \rightarrow 0} \frac{2y \exp(-x^{-2})}{y^2 + \exp(-2x^{-2})} &= \frac{2y^2}{y^2 + \exp(2\ln(y))} \\ &= \frac{2y^2}{y^2 + y^2} = 1 \neq 0 = f(0,0) \end{aligned}$$

Therefore, f is not continuous at $(0,0)$. $\square$

4 Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by

$$f(x) = \frac{x^3 y}{x^4 + y^2}$$

if $(x,y) \neq (0,0)$ and $f(0,0) = 0$. Show that f is Gâteaux differentiable at the origin but not Fréchet differentiable.

Proof: Let $d = (d_1, d_2)$, then

$$\begin{aligned} \frac{f(0+td) - f(0)}{t} &= \frac{(td_1)^3 td_2}{t((td_1)^4 + (td_2)^2)} \\ &= \frac{t^4 d_1^3 d_2}{t^3(t^2 d_1^4 + d_2^2)} \\ &= \frac{td_1^3 d_2}{t^2 d_1^4 + d_2^2} \\ &\rightarrow 0 \text{ as } t \rightarrow 0 \text{ when } d_2 \neq 0 \end{aligned}$$

When $d_2 = 0$, we have that

$$\begin{aligned} \frac{td_1^3 d_2}{t^2 d_1^4 + d_2^2} &= \frac{0}{t^2 d_1^4} \\ &\rightarrow 0 \text{ as } t \rightarrow 0 \end{aligned}$$

Therefore, f has directional derivatives at $(0,0)$ in all directions d and the directional derivative is linear in d at $(0,0)$, so f is Gâteaux differentiable at $(0,0)$. To see that f is not Fréchet differentiable at $(0,0)$, consider $f'(0, (1,0)) = 0 = f'(0, (0,1))$. But

$$\begin{aligned}
\lim_{\|h\| \rightarrow 0} \frac{|f(0+h) - f(0) - \langle (f'(0, (1,0)), f'(0, (0,1))); h \rangle|}{\|h\|} &= \lim_{\|h\| \rightarrow 0} \frac{|f(h)|}{\|h\|} = \lim_{\|h\| \rightarrow 0} \frac{|f(h)|}{\|h\|} \\
&= \lim_{\|h\| \rightarrow 0} \frac{|((h_1)^3 h_2)/((h_1)^4 + (h_2)^2)|}{\sqrt{(h_1)^2 + (h_2)^2}} \\
&= \lim_{\|h\| \rightarrow 0} \frac{|h_1^3 h_2|}{|h_1^4 + h_2^2| \sqrt{h_1^2 + h_2^2}} \\
&= \lim_{\|h\| \rightarrow 0} \frac{|h_1^3 h_2|}{|h_1^2 + h_1^2 + h_2^2| \sqrt{h_1^2 + h_2^2}} \\
&= \lim_{\|h\| \rightarrow 0} \frac{|h_1^3 h_2|}{|h_1^2| + 2\sqrt{h_1^2 + h_2^2}} \\
&= \lim_{\|h\| \rightarrow 0} \frac{|h_1 h_2|}{2\sqrt{h_1^2 + h_2^2}}
\end{aligned}$$

Taking $h_1 = h_2$, we have

$$\lim_{\|h\| \rightarrow 0} \frac{|2h_1|}{2\sqrt{2}h_1} = \frac{1}{\sqrt{2}} \neq 0$$

Therefore, f is not Fréchet differentiable at $(0,0)$. $\square$